Session 1 - Welcome to the IT World Explain careers in IT (developer, tester, designer, data analyst, AI engineer). Discuss product vs service companies; real-world examples (Google, Zoho, TCS). Overview of how teams collaborate using GitHub, Slack, Jira. Trainer Demo: Show an open-source GitHub project (e.g., "Awesome-Java") and walk through contributors' roles. AI Tool Tip: Ask ChatGPT to list top emerging tech jobs (to discuss in class).

TASKS:

1. Name 5 IT job roles and write one line about what each role does.

- **Software Developer**: Designs, codes, and maintains computer programs and applications.
- **Software Tester (QA Engineer):** Tests software applications to find bugs, errors, and defects, ensuring the product works correctly before it goes live.
- **UI/UX Designer:** Creates the visual layout and user experience flows to ensure applications are intuitive and user-friendly.
- **Data Analyst:** Collects, processes, and analyzes large datasets to help organizations make data-driven business decisions.
- **Cloud Engineer:** Manages, maintains, and deploys applications and infrastructure on cloud platforms like AWS, Azure, or Google Cloud.
- **AI Engineer:** Builds, trains, and deploys machine learning models and artificial intelligence
- **Cybersecurity Specialist:** Protects an organization's networks and data from cyber threats, attacks, and unauthorized access.

2. List 3 product companies and 3 service companies with one example of work each does.

PRODUCT COMPANIES:

1. Apple – Manufactures consumer electronics and operating systems.

Example of work: Designs and sells the Apple Vision Pro headset.

2. Google – Develops internet services and software products.

Example of work: Operates the Google Search engine.

3. Microsoft – Creates operating systems and productivity software.

Example of work: Provides Microsoft Office applications like Word, Excel, and PowerPoint.

4. Cisco Systems – Manufactures networking hardware, telecom equipment, and software products.

Example of work: Builds Cisco Catalyst Network Switches for enterprise networks.

SERVICE COMPANIES:

1. Tata Consultancy Services (TCS) – Provides IT services and business solutions**.**

Example of work: Develops and maintains custom supply-chain management applications for clients.

2. Infosys – Provides digital transformation, cloud, and AI services.

Example of work: Helps businesses migrate their systems to the cloud.

3. Accenture – Provides strategy, technology, and consulting services.

Example of work: Helps organizations improve business processes and digital infrastructure.

4. Wipro: Provides cybersecurity services to monitor and protect a client's network from threats.

3. Write the full forms and one-line use of: GitHub, Slack, Jira, IDE, API.

- **GitHub:** Git Hosting Service (commonly called GitHub):

  Use: Used to store, manage, and collaborate on code projects.

- **Slack:** Searchable Log of All Conversation and Knowledge

  Use: A tool used for team messaging, file sharing, collaboration and communication.

- **Jira:** Jira

  Use: Used for project management and issue tracking.

- **IDE:** Integrated Development Environment

  Use: Used to write, edit, test, and debug software code.

- **API:** Application Programming Interface

  Use: Used to allow different software applications to communicate with each other.

4. Name any 5 programming languages and mention one popular use case for each.

Python: Used for data analysis and building machine learning models.

Java: Commonly used for building Android applications and enterprise-level backend software systems.

JavaScript: web development, used to create interactive elements on websites.

C++: Game development and high-performance system software.

PHP: Primarily used for server-side scripting to power dynamic websites and web applications.

Swift: The primary language developed by Apple for building iOS and macOS applications.

5. What is the difference between a developer and a tester? Answer in 2 lines.

Developer:

Writes and maintains software code.

Develops features and application functionality.

Fixes coding issues and implements improvements.

Creates the product.

Focuses on building the software product.

Tester:

Tests software to find bugs and defects.

Verifies that features work as expected.

Reports issues and validates fixes.

Ensures the product is reliable and user-friendly.

Focuses on ensuring software quality.